Quiz 4

Name: _____

AI 501: Mathematics for AI | Duration: 20 min | Student ID: _____
Total: 10 marks

Question 1 (4 marks)

State whether each of the following statements is **True** or **False**. If **False**, give a brief counter-statement.

(a) A wide matrix $A \in \mathbf{R}^{m \times n}$ with $m < n$ can have a left inverse.

(b) If $Ax = b$ has a solution and A has a left inverse, that solution is unique.

(c) A system $Ax = b$ with more equations than unknowns ($m > n$) is guaranteed to have at least one solution.

(d) The pseudo-inverse $A^\dagger$ can be used to estimate x even when $Ax = b$ has no exact solution.

Solution:

(a) **False.** A wide matrix has more columns than rows, so its columns cannot all be independent. Therefore, it cannot have a left inverse.

For example,

$$A = \begin{pmatrix} 1 & 0 \end{pmatrix}.$$

There is no matrix X such that $XA = I_2$.

(b) **True.** Let C be a left inverse of A, so $CA = I$. If x and $\tilde{x}$ are both solutions, then

$$Ax = b, \qquad A\tilde{x} = b.$$

Multiplying both equations by C gives

$$x = CAx = Cb$$

and

$$\tilde{x} = CA\tilde{x} = Cb.$$

Therefore,

$$x = \tilde{x},$$

so the solution is **unique**.

(c) **False.** Having more equations than unknowns does not guarantee that a solution exists.

For example,

$$x = 1, \qquad x = 2$$

cannot both be true. Therefore, this system has **no solution**.

(d) **True.** If $Ax = b$ has no exact solution, the pseudo-inverse can be used to find the **best approximate solution** (least-squares solution).

Question 2 (2 marks)

For each system, write A and b, classify it as under-determined, square, or over-determined, and state whether it has no solution, a unique solution, or infinitely many solutions.

(a) $x_1 - x_2 = 2, \quad x_1 + x_2 = 4, \quad 3x_1 - x_2 = 8.$

(b) $x_1 + x_2 + x_3 = 2, \quad 2x_1 + 2x_2 + 2x_3 = 5.$

Solution:

(a)

$$A = \begin{pmatrix} 1 & -1 \\ 1 & 1 \\ 3 & -1 \end{pmatrix}, \qquad b = \begin{pmatrix} 2 \\ 4 \\ 8 \end{pmatrix}.$$

A is 3×2 ($m > n$), so the system is **over-determined**.

Adding the first two equations gives

$$2x_1 = 6,$$

so

$$x_1 = 3, \qquad x_2 = 1.$$

Check the third equation:

$$3(3) - 1 = 8 \quad \checkmark$$

Therefore, the system has a **unique solution**:

$$x = \begin{pmatrix} 3 \\ 1 \end{pmatrix}.$$

(b)

$$A = \begin{pmatrix} 1 & 1 & 1 \\ 2 & 2 & 2 \end{pmatrix}, \qquad b = \begin{pmatrix} 2 \\ 5 \end{pmatrix}.$$

A is 2×3 ($m < n$), so the system is **under-determined**.

Dividing the second equation by 2 gives

$$x_1 + x_2 + x_3 = 2.5.$$

But the first equation says

$$x_1 + x_2 + x_3 = 2.$$

These two equations contradict each other. Therefore, the system has **no solution**.

Question 3 (2 marks)

Let $u = (2, -1, 1)^T$ and $v = (1, 0, 3)^T$. Compute the outer product $A = uv^T$. What is the rank of the matrix A?

Solution:

$$A = uv^T = \begin{pmatrix} 2 \\ -1 \\ 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 3 \end{pmatrix} = \begin{pmatrix} 2 & 0 & 6 \\ -1 & 0 & -3 \\ 1 & 0 & 3 \end{pmatrix}.$$

Every row is a multiple of

$$(1, 0, 3).$$

Therefore, there is only one independent row, so

$$\text{rank}(A) = 1.$$

In general, the outer product of two nonzero vectors has rank 1.

Question 4 (2 marks)

A linear classifier scores a feature vector $x \in \mathbb{R}^3$ for two classes using $z = Wx$, where $W = \begin{pmatrix} 2 & -1 & 1 \\ 1 & 0 & -1 \end{pmatrix}$, $x = \begin{pmatrix} 1 \\ 3 \\ 2 \end{pmatrix}$. Using $z = Wx$, state which class score is higher.

Solution:

$$z = Wx = \begin{pmatrix} (2)(1) + (-1)(3) + (1)(2) \\ (1)(1) + (0)(3) + (-1)(2) \end{pmatrix} = \begin{pmatrix} 1 \\ -1 \end{pmatrix}.$$

Since

$$1 > -1,$$

Class 1's score is higher than Class 2's score.

Therefore, the classifier predicts **Class 1**.